

Uncertainty II: Insurance and Risk Diversification

JEB104 Microeconomics I
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Uncertainty II: Insurance and Risk Diversification

Insurance
States-of-the-World Approach to Insurance
Risk Diversification and Risk Pooling

Actuarially Fair Insurance
Actuarially Unfair Insurance

Insurance Contract

Risk averse individuals would like to reduce risk when facing a *bad* outcome with some positive probability.

Insurance contract

- Insurance premium p : consumer pays it **before** knowing the outcome, price of insurance
- Insurance benefit b : insured consumer is entitled to it if she ends up facing **only** the “bad” outcome
- Insurance premium can be defined as: $p(b) = \gamma b$, $\gamma \in (0, 1)$

Previously

- Introduction to uncertainty
- Everyday we make choices which have uncertain outcomes.
- In many situations, taking risks is unavoidable.
- Even though driving your car or eating a meal at a restaurant subjects you to some uncertainty about what will actually happen

Now: Insurance

- People are mostly risk averse and are generally willing to sacrifice something to reduce these risks
(risk averse people have positive risk premium)
- One way to reduce the risk is to buy insurance.

Example: Owning a house, it will burn down with a probability of 0.1.

- Bad outcome that it burns down ($p = 0.1$)
- Good outcome is that it will not burn down ($p = 0.9$).
- I can get insured – pay some insurance premium p and get insurance benefit b in case it really burns down. So the amount b is actually insured amount of your house.

Notice that the definition $p(b) = \gamma b$ means that the price of insurance p is a share of insurance benefit (insured amount). So γ is insurance paid per dollar of insured amount.

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Insurance Contract

Risk-averse house-owner:

- house and land worth \$100,000
- land worth \$20,000
- probability that house burns down is 0.1

No-insurance payoff:

- good state: $x_G = e_G = \$100,000$
- bad state: $x_B = e_B = \$20,000$
- $EU_{\text{NOT INSURED}} = 0.9u(x_G) + 0.1u(x_B) = 0.9u(100,000) + 0.1u(20,000)$

Insurance payoff:

- good state: $x_G = e_G - p = \$100,000 - p$
- bad state: $x_B = e_B + b - p = \$20,000 + b - p$
- $EU_{\text{INSURED}} = 0.9u(100,000 - p) + 0.1u(20,000 + b - p)$

What insurance contract (p, b) would the risk-averse house-owner accept?

What Contracts Do Insurance Companies Offer?

“Actuarially fair” insurance markets

- competitive insurance industry
- each insurance company has full information about the risks that individuals face
- risks are not “correlated” across individuals, i.e. in our example 10% of insured owners face the bad outcome, while 90% face the good outcome

N house-owners are insured and pay insurance p

Insurance company receives $N \cdot p$ in revenues and incur $0.1 \cdot N \cdot b$ in costs.

“Actuarially fair” insurance: revenues = expected costs (insurance company)

- given example: $Np = 0.1Nb \rightarrow p = 0.1b$
- generally: $p = q \cdot b$, where q is the probability of the bad outcome
- if $p = \gamma b$, then $\gamma = q$

Here's a more specific example

- House and land worth 100 thousands in total
- But land has some value as well (20 thousands in this case)
- So if the house burns down, the house-owner will still have the land which is worth 20 thousands.
- We denote a wealth a consumer really has in different states of the world as x_G and x_B , while e_G and e_B denote outcomes that occur in different states of the world.
- In the case of no insurance, $x_G = e_G$ and $x_B = e_B$.

Obviously the consumer aims to maximize expected utility

- But he can reach the maximized expected utility only if he is offered an optimal contract.
- This is not always true.
- Generally, agent will accept any insurance contract for which

$$EU_{\text{INSURED}} > EU_{\text{NOT INSURED}}$$

$$0.9(100,000 - p) + 0.1u(20,000 + b - p) > 0.9u(100,000) + 0.1u(20,000)$$

Here, we will look at what contracts can be offered by insurance companies. We firstly look at “actuarially fair” insurance markets.

Risks are not correlated $\Rightarrow$ risks are independent

- This means that it is purely random whose house burns down.
- And that there won't be a fire that destroys the whole city.
- floods – not independent
- lightning striking the house – (more likely) independent, but of course if there is a big storm, more houses could be affected.

N people are insured and each pays insurance premium p

- So the total amount the insurance company receives is Np .

10% of these people lose their house and get insurance benefit b

- Therefore insurance company pays out $0.1Nb$ in total.

Actuarially fair insurance means that the insurance company breaks even:

$$Np = 0.1Nb \Rightarrow p = 0.1b$$

p - what insurance company gets on average, qb - what it expects to pay on average

If insurance is quoted per dollar insured, then the actuarially fair insurance rate γ equals the probability of the bad outcome q .

What Contracts Do Insurance Companies Offer?

“Actuarially fair” insurance

- Insurance companies make zero profit.
- It reduces the risk a consumer faces without changing the expected value of the gamble for which the consumer buys insurance.
- In expectation, consumer pays an amount equal to what she receives back.

Good state: $x_G = e_G - p = e_G - qb$

Bad state: $x_B = e_B + b - p = e_B + b - qb$

$$EV_{\text{INSURED}} = EV_{\text{NOT INSURED}}$$

Notice that we assume that insurance companies do not have other costs (I know it is unrealistic).

Remember: q is the probability of the bad outcome

$$\begin{aligned} EV_{\text{INSURED}} &= (1 - q)(e_G - p) + q(e_B + b - p) \\ &= (1 - q)(e_G - qb) + q(e_B + b - qb) \quad \text{using } p = qb \\ &= (1 - q)e_G - (1 - q)qb + qe_B + qb - q^2b \\ &= (1 - q)e_G + qe_B = EV_{\text{NOT INSURED}} \end{aligned}$$

Note: The last equality holds because $qb - q^2b = qb(1 - q)$

So actuarially fair insurance contract does not change your expected wealth.

What Contracts Do Insurance Companies Offer?

An insurance company offers 3 actuarially fair insurance contracts to the house-owner:

- $(p_1; b_1) = (5,000; 50,000)$
- $(p_2; b_2) = (7,000; 70,000)$
- $(p_3; b_3) = (8,000; 80,000)$

Which one would the house-owner prefer?

No insurance: $EV = 0.9 \cdot 100 + 0.1 \cdot 20 = 92$

Insurance contracts:

1. $x_G^1 = 100 - 5 = 95$, $x_B^1 = 20 + 50 - 5 = 65$, $EV_1 = 0.9 \cdot 95 + 0.1 \cdot 65 = 92$
2. $x_G^2 = 100 - 7 = 93$, $x_B^2 = 20 + 70 - 7 = 83$, $EV_2 = 0.9 \cdot 93 + 0.1 \cdot 83 = 92$
3. $x_G^3 = 100 - 8 = 92$, $x_B^3 = 20 + 80 - 8 = 92$, $EV_3 = 0.9 \cdot 92 + 0.1 \cdot 92 = 92$

So there are three policies to choose from, all satisfying $p = qb = 0.1b$

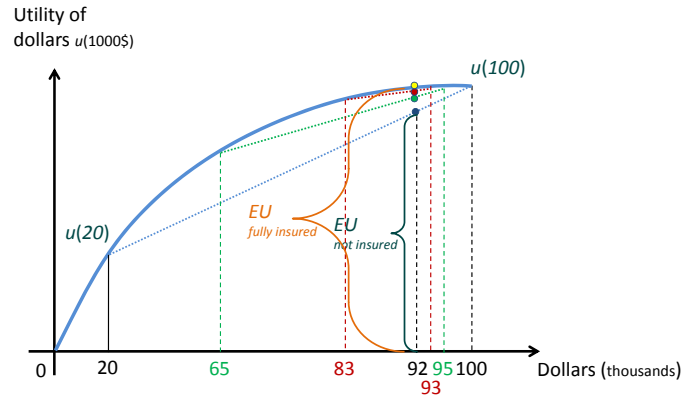
- 1 and 2 cover only partially but cost less ($x_G > x_B$)
- 3 is full insurance, i.e. fully insures the loss of 80,000 (so $x_G = x_B$)

Let's compare the three policies and their expected wealths first.

- All three bring the same expected wealth $\Rightarrow$ A risk neutral consumer would be indifferent between them.
- Can we say something about a risk averse? Look at the picture on the next slide.

Which Contract Would the House-Owner Prefer?

Case 1: green, Case 2: red, Case 3: orange



Full insurance brings risk-averse individual the highest expected utility.

When only money matters, risk-averse individual always chooses full insurance, if it is an actuarially fair contract, hence $b = x_G - x_B$.

- The picture shows the alternative insurance contracts and no insurance (black).
- We have calculated x_G , x_B and expected wealths for all the contracts in the previous slide.
- Now, we will depict expected utility for all the contracts and check which expected utility is the largest.
- Small circles represent EUs for all the contracts.
- The largest EU is represented by the yellow circle which lies on the utility function. In this case, the individual is fully insured, he always has 92,000 irrespective of the state of the world, so $EU = u(92,000)$.

So if you get an actuarially fair offer, you will always choose to insure the full amount if you are risk averse.

Theoretically, which option would be chosen by a risk-loving individual? Will a risk-loving individual get insured? Think about it.

Actuarially Unfair Insurance

Actuarially “unfair” insurance

- Insurance companies make positive profit.
- Expected value falls as an individual insures more heavily.

Insurance company:

$$\text{Revenues} = N \cdot p > q \cdot N \cdot b = \text{Costs}$$

Insured individual:

$$EV_{\text{INSURED}} < EV_{\text{NOT INSURED}}$$

In real life, insurance companies will offer actuarially unfair contracts.

- Operating insurance company incurs other expenses.
- Insurance companies need to make profit for various reasons.
- Also they need to have a reason to insure you.

Risk and expected wealth – the lower risk (the lower difference between x_G and x_B), the lower expected wealth. Look at the contracts on the next slide.

What Contracts Do Insurance Companies Offer?

An insurance company offers 3 actuarially unfair insurance contracts to the house-owner:

- $(p_1; b_1) = (7, 000; 50, 000)$
- $(p_2; b_2) = (10, 000; 70, 000)$
- $(p_3; b_3) = (14, 000; 80, 000)$

Which one would the house-owner prefer?

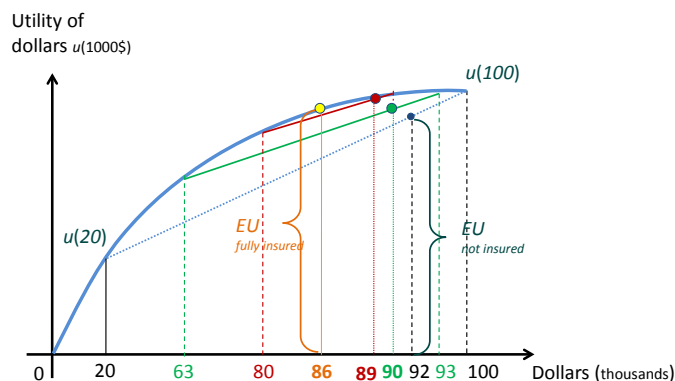
No insurance: $EV = 0.9 \cdot 100 + 0.1 \cdot 20 = 92$

Insurance contracts:

1. $x_G^1 = 100 - 7 = 93$, $x_B^1 = 20 + 50 - 7 = 63$, $EV_1 = 0.9 \cdot 93 + 0.1 \cdot 63 = 90$
2. $x_G^2 = 100 - 10 = 90$, $x_B^2 = 20 + 70 - 10 = 80$,
 $EV_2 = 0.9 \cdot 90 + 0.1 \cdot 80 = 89$
3. $x_G^3 = 100 - 14 = 86$, $x_B^3 = 20 + 80 - 14 = 86$,
 $EV_3 = 0.9 \cdot 86 + 0.1 \cdot 86 = 86$

Which Contract Would the House-Owner Prefer?

Case 1: green, Case 2: red, Case 3: orange



$$EU_{\text{red}} > EU_{\text{orange}} = \text{full insurance} > EU_{\text{green}} > EU_{\text{no insurance}}$$

Full insurance **DOES NOT** always bring risk-averse individual the highest EU!

In the previous case, $\gamma = 0.1$ (remember that it represents the price of 1 dollar insured), now it is higher. Benefits are now the same as in previous contracts, but premiums (insurance prices) are higher.

Note that the γ differs between the three contracts:

$$\gamma_1 = \frac{7}{50} = 0.140, \gamma_2 = \frac{1}{7} = \frac{7}{49} = 0.142, \gamma_3 = \frac{14}{80} = \frac{7}{40} = 0.175$$

i.e. γ increases with greater benefit, it is relatively more costly to insure more heavily

So expected wealth decreases with greater benefit.

We also know that the decision is not based on expected wealth only

- If it would be so, consumer would choose no insurance.

Let's have a look at the next slide to see what are expected utility levels.

We can see that the red combination gives consumer the highest expected utility (i.e. insuring 70,000 and paying the premium of 10,000).

Full insurance $EU = u(86,000)$ does not bring the highest expected utility in this case.

$$EU_{\text{FULLY INSURED}} = u(86,000) < EU_{\text{NOT FULLY INSURED}} = 0.9u(90,000) + 0.1u(80,000)$$

When the red (case 2, optimal) and the orange (case 3, full insurance) cases are compared, moving from the red to the orange case, it means that you have to pay additional 4,000 for sure to get 10,000 in benefit with probability 0.1.

So if actuarially unfair contracts are offered, full insurance is not necessarily optimal! It will always depend on contracts offered and on the shape of a utility function.

Insurance: General Conclusion

While we found that risk averse individuals will choose to fully insure when insurance markets are actuarially fair, this is not the case when insurance markets are actuarially unfair.

No insurance company could stay in business if it offered fair insurance.

Risk-neutral people will never buy unfair insurance. If people are sufficiently risk averse, they will buy unfair insurance.

Suppose that **only full insurance contracts** are offered, i.e. $b = e_G - e_B$:

- What is the maximum premium p individual is willing to pay for the insurance?

$$u(x_G) = u(e_G - p) = EU_{\text{not insured}} \iff e_G - p = x_{CE_{\text{not insured}}}$$

$$p = e_G - x_{CE_{\text{not insured}}}$$

- The more risk averse individuals $\rightarrow$ lower $x_{CE_{\text{not insured}}}$ $\rightarrow$ the higher maximum premium they are willing to pay

Insurance: Problems

- Large-scale disasters
- Adverse-selection: only “worst” customers may end up buying an insurance policy
- Moral hazard: insured customers are less willing to take steps to avoid losses

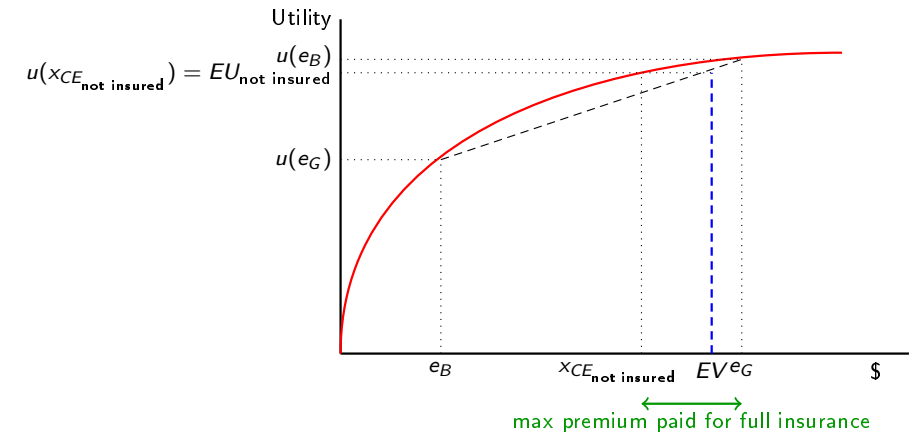
Insurance company cannot afford to offer fair insurance

- Besides covering claims, insurance companies must also maintain records, collect premiums, investigate fraud, so they have many administrative costs.

Risk neutral individuals: $EU = u(EV)$, so they choose the contract giving the highest expected wealth. If only actuarially unfair insurance contracts are offered, not being insured brings the highest expected wealth.

Full insurance contracts satisfy:

$$\begin{aligned} x_G &= x_B \\ e_G - p &= e_B - p + b \\ \text{So } b &= e_G - e_B. \end{aligned}$$



Large-scale disasters

- such as hurricanes or wars;
- may result in such large losses that the insurance company would go bankrupt before it could pay all the claims;
- rare and unpredictable – not floods, nuclear power plant accidents.

Adverse-selection

- Only “worst” customers may end up buying an insurance policy.
- Insurance company has to find a way how to control who buys insurance.
- Screening – young drivers pay more for car insurance (more risky individuals having generally more car accidents)

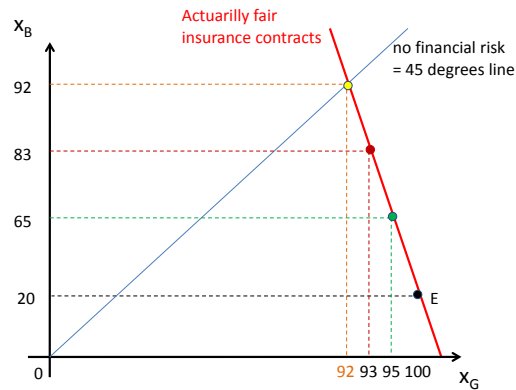
Moral hazard

- Insured customers are less willing to take steps to avoid losses.
- Insured drivers may not be so safe when driving.

Insurance in Different States of the World

Consumption in good state x_G and in bad state x_B

Endowment point: $e_G = 100$, $e_B = 20$



Actuarially fair contracts: $(1 - q)x_G^{AF} + qx_B^{AF} = (1 - q)e_G + qe_B$

Slope: $-\frac{1-q}{q}$

Indifference Curves

Indifference curves: constant expected utility, convex (risk aversion)

$$EU = (1 - q)u(x_G) + qu(x_B)$$

Derive MRS :

Constant $EU \Rightarrow dEU = 0$

$$dEU = (1 - q)MU_{x_G} dx_G + qMU_{x_B} dx_B = 0$$

$$MRS_{BG} = \frac{dx_B}{dx_G} = -\frac{(1 - q)MU_{x_G}}{qMU_{x_B}}$$

States of the world approach to uncertainty

- More general model
- Bears strong resemblance to the model of consumer choice in a two-good world when the consumer is endowed with different amounts of each good but has no other source of income.

The 45-degree line – no financial risk:

- does not matter which state of the world occurs, $x_B = x_G$
- equivalent to full insurance
- Below: policies under which consumers do not fully insure and thus are left with less consumption in the “bad” state than in the “good” state
- Above: “overinsurance” under which the consumer ends up with more consumption in the “bad” state than in the “good” state.

Red line:

- the line of actuarially fair insurance contracts;
- the “budget” line for our consumer who can choose one of the contracts;
- endowment point $E = (e_G, e_B)$, which is the point, where consumer does not buy any insurance, must be available;
- a part of the BC below E does not have any economic meaning.

The slope follows directly from the equation in the slide:

$$\begin{aligned} EV_{\text{insurance}} &= EV_{\text{no insurance}} \\ (1 - q)x_G^{AF} + qx_B^{AF} &= (1 - q)e_G + qe_B \\ (1 - q)(e_G - p^*) + q(e_B - p^* + b^*) &= (1 - q)e_G + qe_B \\ (1 - q)(e_G - p^*) + q(e_B - p^* + \frac{1}{q}p^*) &= (1 - q)e_G + qe_B \\ x_B^{AF} &= -\frac{1 - q}{q}x_G^{AF} + \frac{qe_B + e_G(1 - q)}{q} \end{aligned}$$

One dollar paid on insurance premium gives you in total $\frac{1-q}{q} = \frac{1}{q} - 1 > 1$ if bad state of the world occurs. For actuarially fair insurance contracts $p = qb$, so $b = p/q$. So if one extra dollar is paid on premium p , this extra dollar gives an additional benefit $1/q > 1$ and -1 paid on premium if bad state occurs.

Above we have introduced “budget” constraint of our consumer in the uncertainty setting. Now we should discuss preferences.

We value contracts the same if the expected utilities we get from these contracts are the same.

Convexity with different risk:

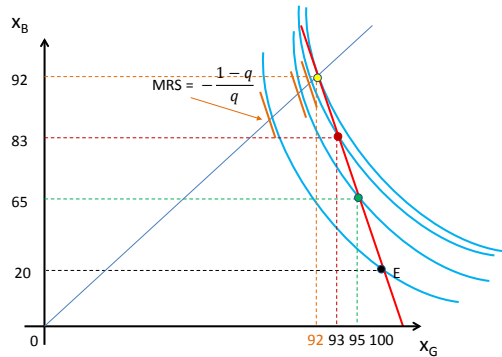
- More now, less in the future, less now and more in the future
- Average is preferred - exactly risk aversion

Interior Optimum

$MRS = \text{slope of budget constraint}$

$$MRS_{BG} = -\frac{(1-q)MU_{x_G}}{qMU_{x_B}} = -\frac{(1-q)}{q}$$

Hence, $MU(x_G) = MU(x_B)$ and $x_G = x_B$, full insurance for risk averse preferences in optimum.



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Actuarially Unfair Insurance

Actuarially unfair insurance: $p = \gamma \cdot b > q \cdot b$

- Slope of the budget line $-\frac{1-\gamma}{\gamma} > -\frac{1-q}{q}$, i.e. budget line is flatter.
- 1 dollar paid on insurance premium gives $\frac{1-\gamma}{\gamma}$ (less than $\frac{1-q}{q}$) if bad state of the world occurs

Interior optimum:

$$MRS_{BG} = -\frac{(1-q)MU_{x_G}}{qMU_{x_B}} = -\frac{1-\gamma}{\underbrace{\gamma}_{< \frac{1-q}{q}}}$$

Hence,

$$\frac{MU_{x_G}}{MU_{x_B}} < 1$$

and $MU_{x_G} < MU_{x_B}$.

For risk averter (diminishing MU), $x_G > x_B \longleftrightarrow$ not fully insured.

So we used a different approach to show that a risk-averse individual will choose full insurance if actuarially fair insurance contracts are available. Before moving to actuarially unfair insurance, let's discuss another point related to difference states of the world.

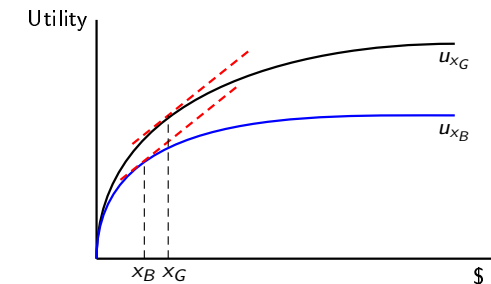
In previous part, we just look at our utility when only money matters

- When our house burns down, we just care about money... but also our memories would burn down
- Sad because of losing what we build, where we lived / not only because of losing wealth

In circumstances where the cause of the bad state is itself undesirable and affects how we evaluate money, we need a more general model of choice in the presence of risk.

$$MRS_{BG} = -\frac{(1-q)MU_{x_G}}{qMU_{x_B}} = -\frac{(1-q)}{q}$$

The above holds, but utility functions in different states can be different, $u'_{x_G}(x_G) = u'_{x_B}(x_B)$ holds, but $u_{x_G}(x) > u_{x_B}(x)$, which may result in $x_G > x_B$.



- You value money more when the state is good, you like the world around.
- But when the state is bad you are depressed, sad, do not need money, you just sit and do not have energy to do anything.
- This may lead to overinsurance, because you will be given more money in a bad state to make the equation hold if optimal insurance is computed by the general model.

Let's get back to the general case in which we assume that the utility function is the same in both states of the world and consider actuarially unfair insurance contracts.

Remind that in general, $p = \gamma b$. In the case of actuarially fair insurance, $\gamma = q$. For actuarially unfair insurance, $\gamma > q$. If insurance contracts are set like according to the general formula, budget line will be linear with slope $-\frac{1-\gamma}{\gamma}$. But notice that in our original example with unfair contracts, γ was not the same for all the contracts offered (budget line would have kink points, or could be even discrete if limited number of contracts are offered).

You need to pay more to get a dollar of insurance back in case of the bad state

- So the budget constraint becomes flatter
- And hence a risk averter will not purchase a full insurance

Risk Diversification

“Don’t put all your eggs in one basket”

Risk diversification may $\uparrow$ expected utility

Example:

- Investor has \$35,000 and invests \$15,000 into risky assets
- Investing into stocks of company A and B, both stocks cost \$1 each
- 50% chance company does well, price of stock increases to \$2
- 50% chance company does poorly, stock is worthless

No diversification (investing into one company):

- 2 possible outcomes \$50,000 and \$20,000

Diversification (\$7,500 invested to each company):

- 4 possible outcomes with 3 different payoffs \$50,000 (25%), \$20,000 (25%), \$35,000 (50%)

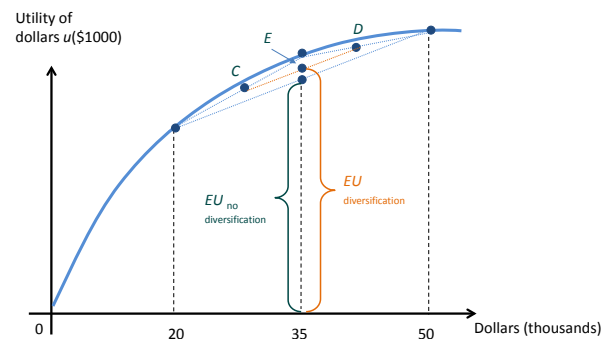
We already did very similar exercise last time, when we were dealing with variance.

In the case of diversification, probability that investor will be earning

- 50,000 is $0.25 = 0.5 \cdot 0.5$ (probability that both companies will be doing well).
- 20,000 is $0.25 = 0.5 \cdot 0.5$ (probability that both companies will be doing poorly).
- 35,000 is $0.5 = 0.25 + 0.25$ (probability that only one of the companies will be doing well).

We will depict the situation in the figure.

Risk Diversification



Diversification in figure: in 2 steps

- C: company B does poorly for sure (50-50 chance A does poorly or well)
- D: company B does well for sure (50-50 chance A does poorly or well)
- E: average of C and D (50-50 chance both A and B do poorly or well)

So we get the diversified outcome “stepwise” from two cases

- A combination of these two cases is then the diversified expected utility

EU from diversification is greater than the EU from no diversification. So, expected wealth is the same ($EV = 35$), but diversification reduces risk, hence increases expected utility.

Risk Pooling

Risk-averse agents can get together and create a pool to share a risk.

Example:

- investment to a fund (you have 50 dollars to invest)
- 50 dollars investment yields with 0.5 probability 100 dollars and with 0.5 probability 0 dollars
- you can invest together with your friend, total return is then divided between both of you.

Investing alone:

- $EV_1 = 0.5 \cdot 100 + 0.5 \cdot 0 = 50$, $EU_1 = 0.5u(100)$
- $\sigma^2 = 0.5(100 - 50)^2 + 0.5(0 - 50)^2 = 2,500$

Investment with your friend:

- $EV_2 = 0.25 \cdot 0 + 0.5 \cdot 50 + 0.25 \cdot 100 = 50$, $EU_2 = 0.5u(50) + 0.25u(100)$
- $\sigma_y^2 = 0.25(0 - 50)^2 + 0.5(50 - 50)^2 + 0.25(100 - 50)^2 = 1,125$

$EU_2 > EU_1 \Rightarrow$ investment with your friend brings you higher EU.

Risk-pooling: $\sigma_{\text{risk pooling}}^2 = \frac{1}{n} \sigma_{\text{no risk pooling}}^2$

- more people in the pool $\Rightarrow$ smaller variance $\Rightarrow$ higher expected utility

Risk Pooling

Example:

- Individual has \$35,000 and faces 0.01 probability of \$10,000 loss each year.
- $EU_{\text{no pooling}} = 0.99u(35,000) + 0.01u(25,000)$
- 1000 individuals
- 10 losses on average, \$100,000 lost each year
- Risks are independent

Risk pooling/spreading: each consumer spreads his risk over all the other consumers

Individuals agree to pay \$100 yearly to compensate others for potential losses.

$$EU_{\text{pooling}} = u(34,900)$$

Risk averse individual: $EU_{\text{pooling}} > EU_{\text{no pooling}}$

In the example we showed that one individual may sell insurance to the other, but why do we have large insurance companies that sell insurance to many people?

$EU_2 > EU_1$ follows from a smaller variance.

Risk averter: $0.5u(50) + 0.25u(100) > 0.5u(100)$, i.e. $u(50) > 0.5u(100)$ (don't forget that the risk averter has a concave utility function).

Independent risks

- The probability that any person incurs a loss does not affect the probability that any other incur losses

100 yearly = average loss

- Some years there can be more losses, some years less = reserves.

Risk averse individual:

$$EU_{\text{pooling}} > EU_{\text{no pooling}}$$

$$u(EV) = u(34,900) > EU_{\text{no pooling}}$$

So it is exactly equivalent to full insurance in actuarially fair insurance market.